

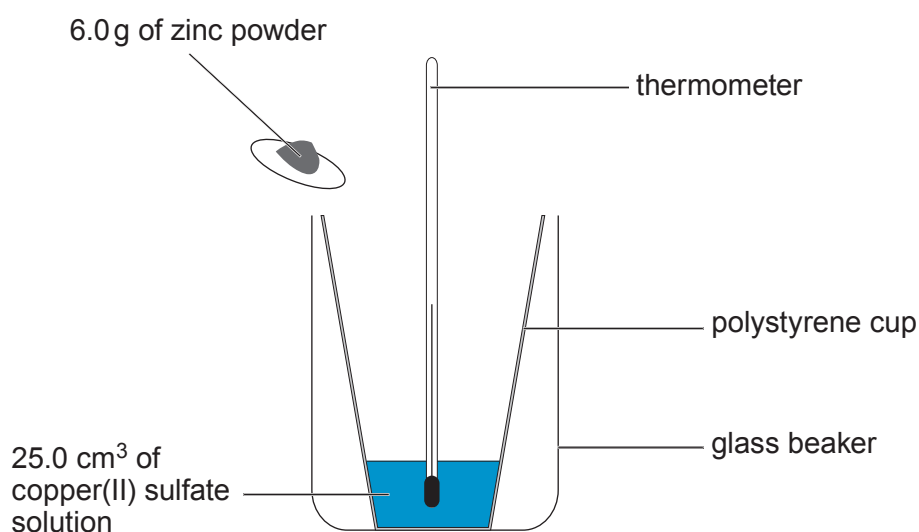
4. (a) The list shows part of the reactivity series.

magnesium
zinc
iron
copper

Underline the correct word(s) in the brackets to complete the following sentence. [1]

The reactivity of metals (**increases** / **stays the same** / **decreases**) down the list.

- (b) Aled investigated the reaction between zinc and excess copper(II) sulfate solution.



Aled put 25.0 cm³ of copper(II) sulfate solution into a polystyrene cup and recorded its temperature. He added 6.0 g of zinc powder to the solution and recorded the temperature every 10 s for 70 s. His results are shown in the table.

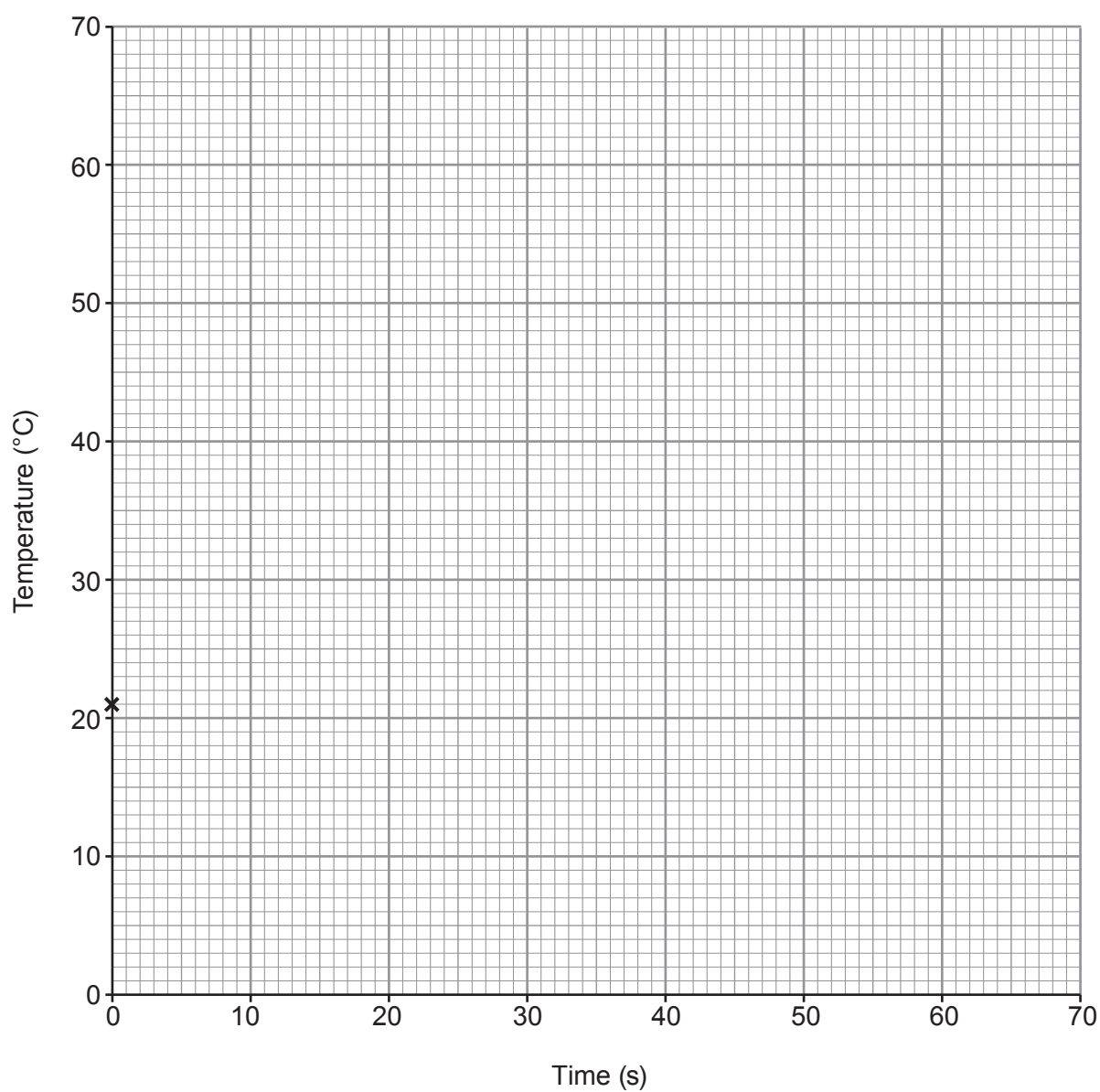
Time (s)	0	10	20	30	40	50	60	70
Temperature (°C)	21	38	50	58	60	59	56	52

- (i) Plot these results on the grid opposite. Draw a suitable line.

The first point has been plotted for you.

[3]





(ii) **Circle** the time it took for **all** the zinc powder to be used up.

[1]

30 s

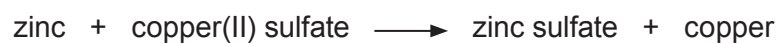
40 s

50 s

70 s



- (iii) The word equation for the reaction between zinc and copper(II) sulfate solution is as follows.



- I. Complete the symbol equation for the reaction. [2]



- II. Underline the type of reaction taking place. [1]

combustion

displacement

neutralisation

addition

